

# NGUYEN VAN MINH

## Backend Developer Intern

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An Nhon Ward, Ho Chi Minh City, Vietnam

## SUMMARY

Final-year Software Engineering student with strong hands-on experience in Java/Spring Boot backend development and ReactJS/Next.js frontend development. Skilled in building RESTful APIs, authentication flows, database-driven applications, and real-time features. Familiar with MySQL, MariaDB, PostgreSQL, MongoDB, Redis, Git, Docker, AWS, and cloud-based deployment. Eager to grow as a Fullstack Developer while expanding practical experience in Cloud Engineering, Python, Go, and AI-powered applications through real-world projects.

## EDUCATION

### Industrial University of Ho Chi Minh City - IUH

Ho Chi Minh City, Vietnam

Major: Software Engineering

Sep 2022 – Present

GPA: 3.2 / 4.0

## SKILLS

### Technical Skills

- **Languages:** Java, JavaScript, TypeScript, Python
- **Frameworks:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Spring AI
- **Databases:** MySQL, MariaDB, PostgreSQL, MongoDB, Redis
- **Tools:** IntelliJ IDEA, Visual Studio Code, Cursor IDE, Maven, Postman, Swagger/OpenAPI
- **DevOps & Cloud:** Docker, Git, GitHub Actions, AWS (IAM, EC2, RDS, DynamoDB, S3, CloudFront)
- **OS:** Windows, Linux

### Soft Skills

- **Problem Solving:** Analytical thinking, debugging, troubleshooting and continuous learning mindset
- **Collaboration:** Teamwork, knowledge sharing, effective communication and intermediate English proficiency
- **Adaptability:** Quickly learning new technologies and tools

## PROJECTS

### Aurora Hotel Management System

Team size: 4 | Role: Backend Developer

- **Auth & RBAC:** Built JWT access/refresh workflow with introspection and invalidation, OAuth2 resource server configuration and 5 roles (Admin / Manager / Staff / Customer / Guest) with 70+ permissions.
- **Hotel Operations:** Developed room/type/category modules, room state tracking, flexible pricing (base/weekend/holiday), media galleries and multi-branch management with branch-level reporting.
- **Booking Engine:** Implemented real-time availability checks, multi-room booking flows, unique booking codes and full modification/cancellation lifecycle handling.
- **Payments:** Integrated VNPay Sandbox with IPN processing, payment status tracking.
- **AI Assistant:** Integrated AI-powered RAG chatbot using Google Gemini, LangChain4j, pgvector semantic search, document upload/chunking and streaming responses.
- **Deployment:** Deploy to AWS EC2 with Docker Hub CI/CD pipelines using GitHub Actions.

**Repository:** [nvminh162/aurora-hotel-management-system](https://github.com/nvminh162/aurora-hotel-management-system)

**Production:** <https://aurora-hotel.nvminh162.id.vn>

### Chatly Social Platform

Team size: 4 | Role: Backend Developer

- **Auth & Security:** Built JWT authentication, refresh tokens, email verification, password reset/change and RBAC admin access.
- **Real-Time Chat:** Developed real-time chat APIs with Spring WebSocket/STOMP, supporting private/group conversations, typing/seen events, edit/recall/delete, forwarding, reactions, polls, pinned messages, mentions, unread counts and search.
- **Group Collaboration:** Implemented group collaboration backend with roles, invite links, join approvals, ownership transfer, notes, reminders and scheduled reminder system messages.
- **Data & Storage:** Designed hybrid PostgreSQL + MongoDB persistence and integrated AWS S3 for media/file storage.
- **Administration:** Built admin and social APIs for dashboard stats, audit logs, reports, users, posts, comments, reactions, saved posts, feeds, stories, reels, conversations, messages and health checks.

**Repository:** [nvminh162/chatly-social-platform](https://github.com/nvminh162/chatly-social-platform)

**Production:** <https://chatly.nvminh162.id.vn>

## Backend - Library Management System

Team size: 1 | Role: Backend Developer

- **Microservices Architecture:** Designed and built a distributed system with 6 independent Spring Boot services (Book, Borrow, Employee, User, Notification, Gateway), registered via Netflix Eureka Discovery and routed through Spring Cloud Gateway.
- **CQRS & Event Sourcing:** Implemented CQRS with Axon Framework and AxonServer — separating Command/Query models, persisting every state change as an immutable event stream with full audit history and point-in-time recovery.
- **Saga & Distributed Transactions:** Engineered a BorrowingSaga orchestrating cross-service distributed transactions - checking book availability, validating employee discipline status, and automatically rolling back all affected services via compensating transactions on failure.
- **Event-Driven Notifications:** Integrated Apache Kafka (KRaft mode) for async communication; implemented @RetryableTopic with exponential back-off 2 multiplier, 4 attempts and Dead Letter Topic (DLT) strategy for fault-tolerant email delivery.
- **Observability:** Integrated Kafka UI (Provectus), RedisInsight and AxonServer dashboard for real-time monitoring of message queues, cache state and event store.

**Repository:** <https://github.com/nvminh162/backend-library-system>

## Nexatech E-commerce Platform

Team size: 4 | Role: Full-stack Developer (Auth, Order, Payment)

- **Security & IAM:** Implemented Keycloak authentication and IAM flows including JWT login/register, token refresh/logout, password recovery and role/group/session management.
- **Order & Checkout:** Developed order and checkout services, idempotency keys, inventory reservation, VNPay/COD payment orchestration, order state machine and timeout-based payment cancellation.
- **Payment Integration:** Integrated VNPay payment service with signed payment URL generation, IPN/return verification, payment lifecycle tracking and COD confirmation.
- **Reliability:** Improved reliability through Kafka event-driven communication, transactional outbox, processed-event deduplication, retry/circuit-breaker configuration, audit logs, snapshots and point-in-time traceability.

**Repository:** <https://github.com/nvminh162-archive/nexatech-ecommerce>

## AWARDS

**Academic Excellence Scholarship - Partial (50%)**

2024 - 2025

Industrial University of Ho Chi Minh City | GPA 3.42/4.0